

Dongfang Zhao, Ph.D.

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Education

University of Washington, Seattle	Computer Science, PostDoc	2017
Illinois Institute of Technology, Chicago	Computer Science, PhD	2015
Emory University, Atlanta, GA	Computer Science, MS	2008
Katholieke Universiteit Leuven, Belgium	Statistics, Diploma	2005

Employment

2023–present	University of Washington ◦ Adjunct Assistant Professor, Allen School of Computer Science & Engineering ◦ Affiliate Faculty of Data Science, eScience Institute ◦ Tenure-track Assistant Professor, Tacoma School of Engineering & Technology
2018–2023	Affiliate Faculty, University of California, Davis
2018–2023	Tenure-track Assistant Professor, University of Nevada, Reno (UNR)
2015	Research Intern, IBM Almaden Research Center, San Jose, CA
2014	Research Intern, Argonne National Laboratory, Chicago, IL
2013	Research Intern, Pacific Northwest National Laboratory, Seattle, WA
2009–2011	Software Developer (full-time), Epic Systems Corporation, Madison, WI
2008–2009	Software Developer (full-time), IDMI Systems, Atlanta, GA

Awards

2022	Federal Research Grant, UNR PI, NSF 2112345, \$255,916 “A DLT Machine Learning Platform for Blockchain Warehousing”
2020	Federal Research Grant, UNR PI, DOE SC0020455, \$200,000 “HPChain: Energy-Efficient Scalable Blockchains for HPC Systems”
2019	Best Paper Award, International Conference on Cloud Computing “BlockLite: Toward Accurate and Efficient Emulation of Public Blockchains in the Cloud”
2018	Best Student Paper Award, IEEE International Conference on Cloud Computing “Toward Cost-Effective Memory Scaling in Clouds: Symbiosis of Virtual and Physical Memory”
2016	Postdoctoral Fellowship, Sloan Foundation, \$155,000

Selected Research Projects

AdaDisk	Distributed agentic system for adaptive ingestion-query scheduling of DiskANN in LLM serving [Github: https://github.com/hpdc/DiskANN] <i>Tags:</i> large language models, agent-oriented programming, distributed systems
Hermes	MySQL plug-in for high-performance homomorphic encryption of vector databases in RAG [Github: https://github.com/hpdc/hermes] <i>Tags:</i> retrieval-augmented generation, vector databases, homomorphic encryption

Selected Publications (Dongfang Zhao was/is the *student's academic advisor)

- [1] *Chengyu Gong, *Gefei Shen, Luanzheng Guo, Nathan Tallent, **Dongfang Zhao**. Order-Preserving Dimension Reduction for Multimodal Semantic Embedding. The 40th Annual AAAI Conference on Artificial Intelligence (AAAI), 2026.
- [2] *Jiuzhou Fu, Luanzheng Guo, Nathan Tallent, **Dongfang Zhao**. ProHD: Projection-Based Hausdorff Distance Approximation. IEEE International Conference on Data Mining (ICDM), 2025.
- [3] *Jiayao Wang, ..., **Dongfang Zhao**. IPBA: Imperceptible Perturbation Backdoor Attack in Federated Self-Supervised Learning. European Conference on Artificial Intelligence (ECAI), 2025.
- [4] *Abdullah Al-Mamun, **Dongfang Zhao**, et al. ZTP: A Scalable and Lightweight Privacy-Preserving Blockchain via Scale-Free Quorums and Geometric Fragmentation. International Conference on Parallel Processing (ICPP), 2025.
- [5] *Gefei Shen, **Dongfang Zhao**. Towards Nonparametric Topological Layers in Neural Networks. The 28th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD), 2024.
- [6] *Muhammad Jahanzeb Khan, ..., **Dongfang Zhao**. Privacy-Preserving Artificial Intelligence on Edge Devices: A Homomorphic Encryption Approach. IEEE International Conference on Web Services (ICWS) 2024: 395-405.
- [7] *Ramasoumya Naraparaju, *Tianyu Zhao, *Yanting Hu, **Dongfang Zhao**, Luanzheng Guo, Nathan R. Tallent. Shifting Between Compute and Memory Bounds: A Compression-Enabled Roofline Model. SC Workshops 2024: 309-316.
- [8] *Olamide Timothy Tawose, Jun Dai, Lei Yang, **Dongfang Zhao**. Toward Efficient Homomorphic Encryption for Outsourced Databases through Parallel Caching. Proceedings of the ACM on Management of Data (SIGMOD), 2023.
- [9] *Xinying Wang, Lipeng Wan, Scott Klasky, **Dongfang Zhao**, Feng Yan: SciLance: Mitigate Load Imbalance for Parallel Scientific Applications in Cloud Environments. International Conference on Cluster Computing (CLUSTER), 2023.
- [10] *Olamide Timothy Tawose, Lei Yang, **Dongfang Zhao**. TopoCommit: A Topological Commit Protocol for Cross-Ledger Transactions in Scientific Computing. International Conference on Cluster Computing (CLUSTER), 2023.
- [11] *Abdullah Al-Mamun, **Dongfang Zhao**. ECHAIN: Securing Electronic Device Provenance through Privacy-Preserving Consortium Blockchains. The 53rd Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN), 2023.
- [12] *Muhammad Jahanzeb Khan, *Olamide Timothy Tawose, Zoey Rui Hu, **Dongfang Zhao**. Exploring the Efficacy of Data-Decoupled Federated Learning for Image Classification and Medical Imaging Analysis. International Workshop on Federated Learning for Distributed Data Mining, co-located with the 29th ACM SIGKDD Conference (KDD), 2023.
- [13] Shuo Chen, Di Li, Bobo Ju, Linhua Jiang, **Dongfang Zhao**. Transparent Object Detection with Simulation Heatmap Guidance and Context Spatial Attention. International Conference on Multimedia Modelling (MMM) (2) 2023: 3-15.
- [14] **Dongfang Zhao**. Cross-Blockchain Transactions: Systems, Protocols, and Topological Theory. In *Handbook on Blockchain*, pp. 157-193. Cham: Springer International Publishing, 2022.
- [15] *Xinying Wang, ..., **Dongfang Zhao**. Memory Scaling of Cloud-based Big Data Systems: A Hybrid Approach. IEEE Transactions on Big Data (TBData). 8(5): 1259-1272 (2022).

- [16] *Abdullah Al-Mamun, Haoting Shen, **Dongfang Zhao**. DEAN: A Lightweight and Resource-efficient Blockchain Protocol for Reliable Edge Computing. The 36th IEEE International Symposium on Parallel and Distributed Processing (IPDPS), 2022.
- [17] *Olamide Timothy Tawose, *Bin Li, Lei Yang, Feng Yan, **Dongfang Zhao**. Topological Modeling and Parallelization of Multidimensional Data on Microelectrode Arrays. The 36th IEEE International Symposium on Parallel and Distributed Processing (IPDPS), 2022.
- [18] *Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**. SciChain: Blockchain-enabled Lightweight and Efficient Data Provenance for Reproducible Scientific Computing. 37th IEEE International Conference on Data Engineering (ICDE), 2021.
- [19] *Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**. BAASH: lightweight, efficient, and reliable blockchain-as-a-service for HPC systems. The International Conference for High Performance Computing, Networking, Storage and Analysis (SC), 2021.
- [20] *Xinying Wang, *Olamide Timothy Tawose, Feng Yan, **Dongfang Zhao**. HDK: Toward High-Performance Deep-Learning-Based Kirchhoff Analysis. The Thirty-Fourth AAAI Conference on Artificial Intelligence (AAAI), 2020.
- [21] *Abdullah Al-Mamun, Jun Dai, Xiaohua Xu, Mohammad Sadoghi, Haoting Shen, **Dongfang Zhao**. Poster DEAN: A Blockchain-Inspired Consensus Protocol Enabling Trustworthy Edge Computing. Network and Distributed System Security Symposium (NDSS). 2020.
- [22] *Abdullah Al-Mamun, ..., **Dongfang Zhao**. Poster Reflector: a fine-grained I/O tracker for HPC systems. The 25th ACM Symposium on Principles and Practice of Parallel Programming (PPoPP), 2020.
- [23] **Dongfang Zhao**, Mohamed Mohamed, Heiko Ludwig. Locality-Aware Scheduling for Containers in Cloud Computing. IEEE Transactions on Cloud Computing (TCC). 8(2): 635-646 (2020).
- [24] *Xinying Wang, *Abdullah Al-Mamun, Feng Yan, **Dongfang Zhao**. Toward Accurate and Efficient Emulation of Public Blockchains in the Cloud. The 12th International Conference on Cloud Computing (CLOUD), 2019.
- [25] Heyang Qin, Syed Zawad, Yanqi Zhou, Lei Yang, **Dongfang Zhao**, Feng Yan. Swift machine learning model serving scheduling: a region based reinforcement learning approach. The International Conf. for High Performance Computing, Networking, Storage and Analysis (SC), 2019.
- [26] *Xinying Wang, Cong Xu, Ke Wang, Feng Yan, **Dongfang Zhao**. Toward Cost-Effective Memory Scaling in Clouds: Symbiosis of Virtual and Physical Memory. The 11th IEEE International Conference on Cloud Computing (IEEE CLOUD), 2018.
- [27] *Jiayao Wang, *Abdullah Al-Mamun, Tonglin Li, Linhua Jiang, **Dongfang Zhao**. Toward Performant and Energy-efficient Queries in Three-tier Wireless Sensor Networks. Proceedings of the 47th International Conference on Parallel Processing (ICPP), 2018.
- [28] Parmita Mehta, Sven Dorkenwald, **Dongfang Zhao**, Tomer Kaftan, Alvin Cheung, Magdalena Balazinska, Ariel Rokem, Andrew J. Connolly, Jacob VanderPlas, Yusra AlSayyad. Comparative Evaluation of Big-Data Systems on Scientific Image Analytics Workloads. VLDB, 2017.
- [29] **Dongfang Zhao**, Kan Qiao, Zhou Zhou, Tonglin Li, Zhihan Lu, Xiaohua Xu. Toward Efficient and Flexible Metadata Indexing of Big Data Systems. IEEE Transactions on Big Data (TBData). 3(1): 107-117 (2017).
- [30] **Dongfang Zhao**, Kan Qiao, Zhou Zhou, Tonglin Li, Xiaobing Zhou, Ioan Raicu. Exploiting multi-cores for efficient interchange of large messages in distributed systems. Concurrency and Computation: Practice and Experience (CCPE). 28(13): 3568-3585 (2016).

- [31] **Dongfang Zhao**, Ke Wang, Kan Qiao, Tonglin Li, Iman Sadooghi, Ioan Raicu. Toward high-performance key-value stores through GPU encoding and locality-aware encoding. *Journal of Parallel and Distributed Computing (JPDC)*. 96: 27-37 (2016).
- [32] **Dongfang Zhao**, Kan Qiao, Jian Yin, Ioan Raicu. Dynamic Virtual Chunks: On Supporting Efficient Accesses to Compressed Scientific Data. *IEEE Transactions on Services Computing (TSC)*. 9(1): 96-109 (2016).
- [33] **Dongfang Zhao**, Ning Liu, Dries Kimpe, Robert B. Ross, Xian-He Sun, Ioan Raicu. Towards Exploring Data-Intensive Scientific Applications at Extreme Scales through Systems and Simulations. *IEEE Transactions on Parallel and Distributed Systems (TPDS)*. 27(6): 1824-1837 (2016).
- [34] **Dongfang Zhao**, NagaPramod Mandagere, Gabriel Alatorre, Mohamed Mohamed, Heiko Ludwig. Toward locality-aware scheduling for containerized cloud services. *IEEE International Conference on Big Data (BigData)*, 2015.
- [35] **Dongfang Zhao**, Kan Qiao, Ioan Raicu. HyCache+: Towards Scalable High-Performance Caching Middleware for Parallel File Systems. *The 14th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGrid)*, 2014.
- [36] **Dongfang Zhao**, Kent Burlingame, Corentin Debains, Pedro Alvarez-Tabio, Ioan Raicu. Towards high-performance and cost-effective distributed storage systems with information dispersal algorithms. *IEEE International Conference on Cluster Computing (CLUSTER)*, 2013.
- [37] **Dongfang Zhao**, Chen Shou, Tanu Malik, Ioan Raicu. Distributed data provenance for large-scale data-intensive computing. *International Conf. on Cluster Computing (CLUSTER)*, 2013.
- [38] Tonglin Li, Xiaobing Zhou, Kevin Brandstatter, **Dongfang Zhao**, Ke Wang, Anupam Rajendran, Zhao Zhang, Ioan Raicu. ZHT: A Light-Weight Reliable Persistent Dynamic Scalable Zero-Hop Distributed Hash Table. *IEEE International Symposium on Parallel and Distributed Processing (IPDPS)*, 2013.
- [39] **Dongfang Zhao**, Li Yang. Incremental Isometric Embedding of High-Dimensional Data Using Connected Neighborhood Graphs. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*. 31(1): 86-98 (2009).
- [40] **Dongfang Zhao**, Li Yang. Incremental Construction of Neighborhood Graphs for Nonlinear Dimensionality Reduction. *International Conference on Pattern Recognition (ICPR)*, 2006.